

Vinay Venkatesh

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EDUCATION

Carnegie Mellon University

Pittsburgh, PA

Bachelor of Science in Statistics/Machine Learning

May 2027

- **Relevant Coursework:** Machine Learning, Computer Vision, Deep Learning, Data Structures and Algorithms, Probability, Data Science, Functional Programming, Linear Algebra, Statistical Visualization
- **Activities:** Data Science Club, Artificial Intelligence Club, Asian Student Association, Phi Delta Theta Fraternity

EXPERIENCE

Software Engineer Intern

May 2026 – Aug. 2026

LinkedIn

Mountain View, CA

- Cut deployment freshness lag for a production ML inference tier from **11 days to under 1 hour** with an autonomous Java/Temporal release-certification pipeline, replacing a 5-step manual release process
- Built **end-to-end canary certification** for a GPU-backed retrieval-and-ranking inference service, forking live production traffic onto H200 baseline-vs-test clusters
- Drove a **statistical validity fix** in a partner team's canary platform, replacing per-request latency diffs with baseline-vs-test **P95 distribution comparison** — redefining pass/fail for every canary on the platform
- Shipped changes across **8 repositories owned by 5+ teams**, including a **~1,000-line proposal** adding operation-scoped authorization grants to LinkedIn's service authorization platform

AI Engineer Intern

Jan. 2026 – Apr. 2026

Wolfe, LLC

Pittsburgh, PA

- Reduced campaign production time **98%, 4 hr to 5 min**, with an AWS pipeline (Lambda, Step Functions, Bedrock) generating marketing content across 5 brands and 3 tiers
- Integrated Triple Whale analytics for **data-driven feedback loops**, using post-campaign metrics to optimize AI prompts and targeting
- Designed **YAML configuration system** enabling brand voice, prompt, and template changes without code changes

Software Engineer Intern

June 2025 – Aug. 2025

Payactiv

Milpitas, CA

- Developed **30+ RESTful API endpoints** for an XMPP-based chat platform supporting **1M+ users**
- Designed and updated PostgreSQL schemas, reducing query latency **15%**

Machine Learning Engineer Intern

Feb. 2023 – Aug. 2024

San Jose State University

San Jose, CA

- Optimized ML pipeline linking 3D facial geometry and DNA, achieving **0.73 AUROC** in face-DNA verification
- Compressed **120K+ vertex meshes into 16-D embeddings** using SpiralNet++ for efficient geometric learning
- Improved verification accuracy **18%** using 41 biologically tested SNPs instead of PCA on 3.8M genomic markers

PROJECTS

CreativeGuard | *Python, LLMs, TypeScript, Full-Stack Development*

- Built an AI marketing-compliance platform end to end, evaluating ad creatives against a **35-rule taxonomy** spanning regulatory, bias, accessibility, and brand safety issues
- Designed LLM detection chains producing rule-level verdicts across **6 compliance categories**, fine-tuning on marketing creative data

ArmVision | *Python, Computer Vision, Tensorflow, Research*

IEEE Publication Link

- Developed a **real-time firearm detection system**, training YOLOv5 on a curated dataset to **93% peak accuracy**
- Evaluated model performance across precision, recall, and latency benchmarks
- Published and presented findings at an IEEE conference

TradeSim | *Node.js, Express.js, RESTful APIs, Fintech, Full-Stack Development*

GitHub Link

- Implemented an order matching engine with partial fills, price-priority execution, and multi-symbol order books
- Built a responsive trading dashboard displaying 6 real-time KPIs behind 9 RESTful API endpoints

SKILLS

Languages: Python, Java, TypeScript, Go, Rust, C#, JavaScript, SQL (Postgres), R

Frameworks & Libraries: React, PyTorch, Angular, Node.js, Tensorflow, pandas, NumPy, Matplotlib

Infrastructure & Tools: AWS, Temporal, Kubernetes, Docker, GitHub, Claude Code, Linux, Postman

Methodologies: Retrieval Augmented Generation, Time-Series Forecasting, Regression, GANs, PCA, Agile, Scrum